Signal Processing Basics for Hydroacoustics

From recorded pressure to frequency content

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Learning objectives

By the end of this unit, you should be able to:

- Interpret a hydrophone time series.
- Explain sampling frequency and Nyquist frequency.
- Distinguish between the time and frequency domains.
- Interpret a frequency spectrum produced using the FFT.
- Explain the purpose of basic signal filtering.
- Interpret a spectrogram.

From hydrophone to digital signal

A hydrophone recording begins as a physical pressure fluctuation in the water.
The recording chain is:

Sound pressure
↓
Hydrophone
↓
Analogue electrical signal
↓
Analogue-to-digital conversion
↓
Numerical time series

Once digitised, the signal can be stored, visualised and analysed using a computer.

Sampling a continuous signal

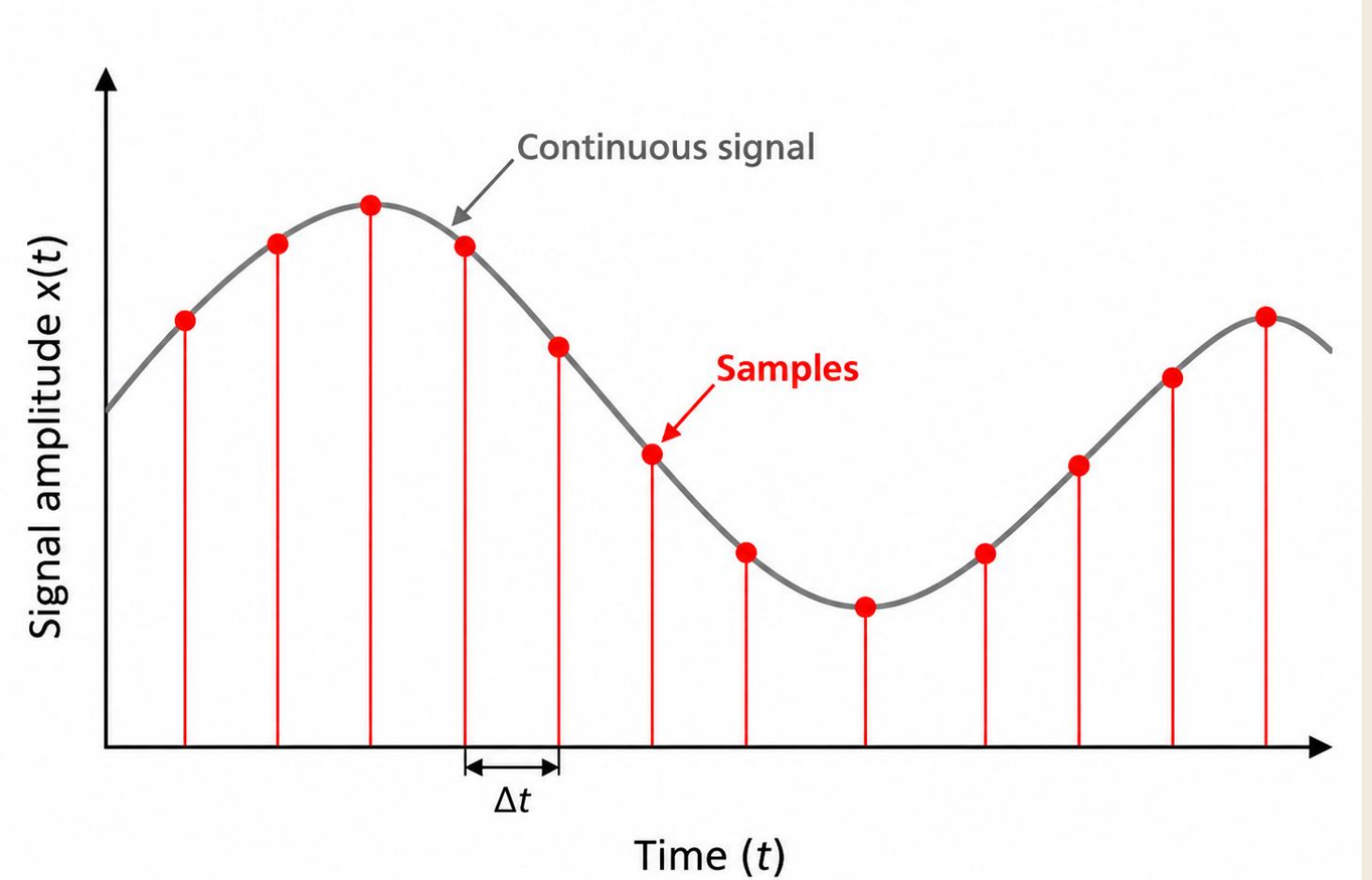
Real underwater sound varies continuously through time.

A digital recorder cannot store every possible instant. Instead, it measures the signal at discrete time intervals. Each measurement is a sample. The sequence of samples forms a digital time series.

Sampling frequency, f_s
Number of samples recorded per second.

Unit: Hertz (Hz)

For example:
48,000 Hz = 48,000 samples per second



Samples are taken at discrete time intervals Δt.

Why does sampling frequency matter?

The sampling frequency determines how rapidly the signal is measured. A higher sampling frequency records more samples per second.

This allows higher-frequency variations in the original signal to be represented. If the sampling frequency is too low, high-frequency information may be represented incorrectly.

This effect is called aliasing.

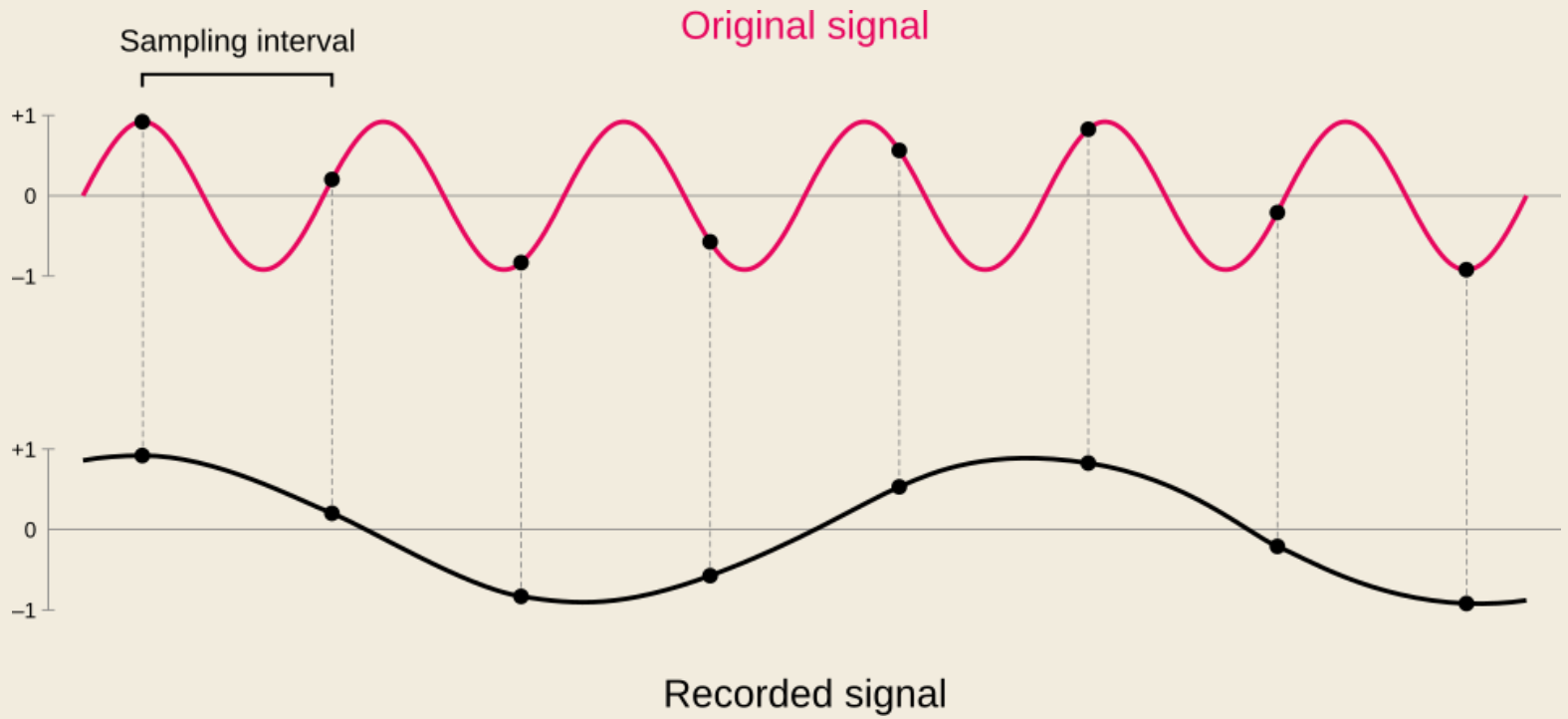
The **Nyquist frequency** is half the sampling frequency:

$$f_N = \frac{f_s}{2}$$

f_N = Nyquist frequency

f_s = sampling frequency

To avoid aliasing, the analogue signal must contain no frequencies above the Nyquist limit before sampling. In practice, anti-aliasing filters are used to attenuate higher frequencies before analogue-to-digital conversion.



Aliasing: When a signal is sampled too slowly, a high-frequency component can appear as a different, lower frequency in the digital recording.

https://commons.wikimedia.org/wiki/File:Signal_aliasing_demonstration.svg

Check — Can we record this sound?

1. f_s = 1000 Hz. What is the Nyquist frequency?

2. Can a 600 Hz signal be correctly represented if f_s = 1000 Hz?

Check — Can we record this sound?

1. f_s = 1000 Hz. What is the Nyquist frequency?

2. Can a 600 Hz signal be correctly represented if f_s = 1000 Hz?

Answers

1. 500 Hz

2. No. The Nyquist frequency is 500 Hz. A 600 Hz tone exceeds this limit and, under ideal sampling without anti-alias filtering, would appear as a 400 Hz tone in the sampled data.
This incorrect representation is called aliasing.

The time domain

A waveform represents a signal in the time domain.

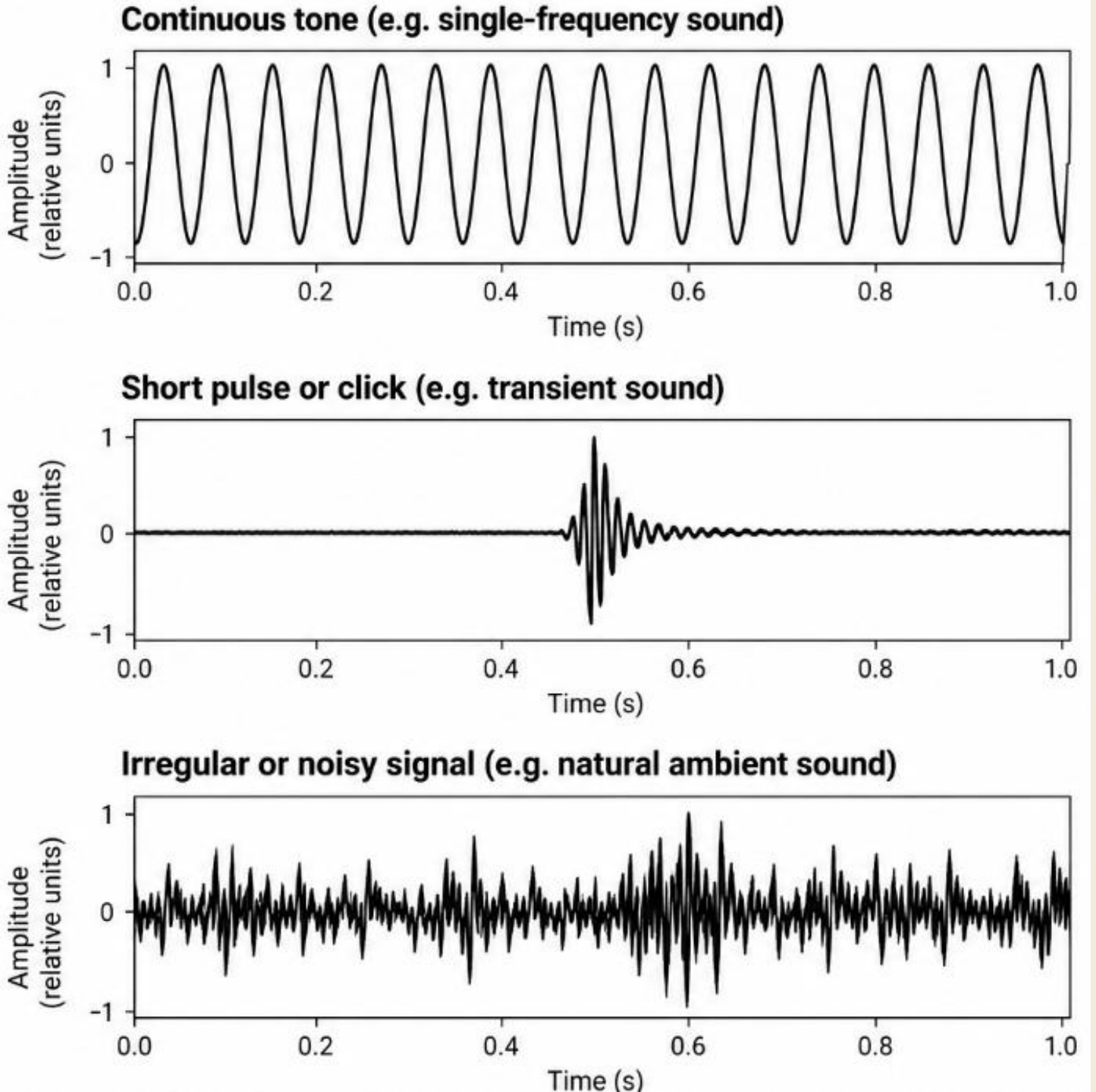
x-axis: time

y-axis: amplitude or pressure

A waveform shows how the recorded signal changes through time. Different signals can produce very different waveform patterns.

Examples:

- continuous tone
- short pulse or click
- irregular or noisy signal

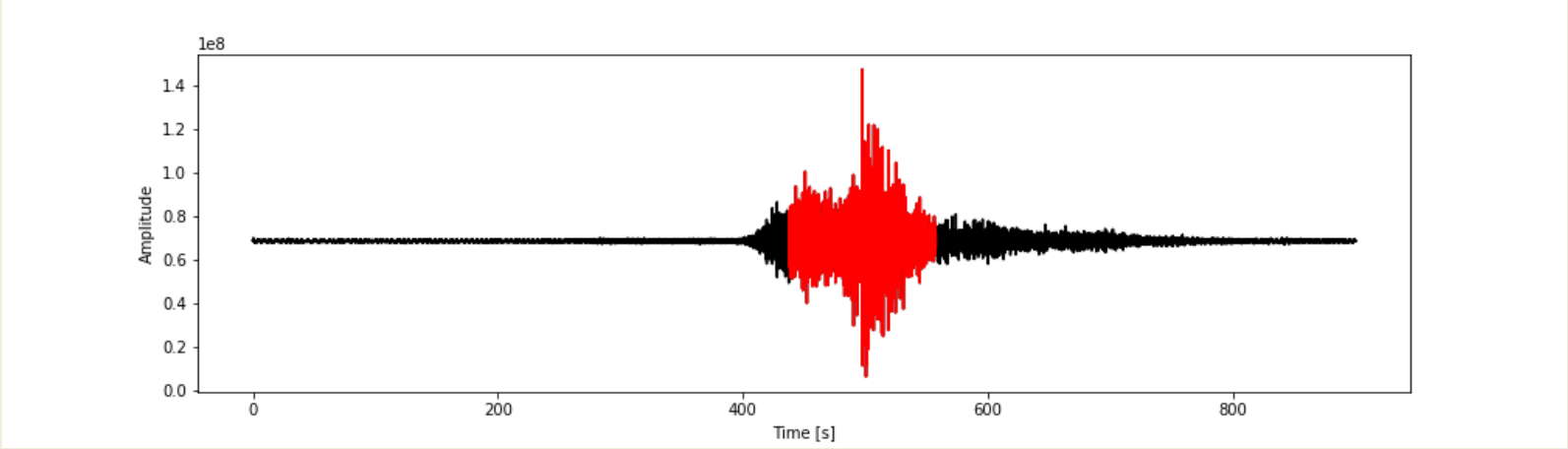


What can a waveform tell us?

A waveform can reveal:

- when a sound occurs
- how long it lasts
- repeated pulses or events
- changes in amplitude
- quiet and active periods

However, the frequency components of a complex sound may be difficult to identify directly from the waveform.



The frequency domain

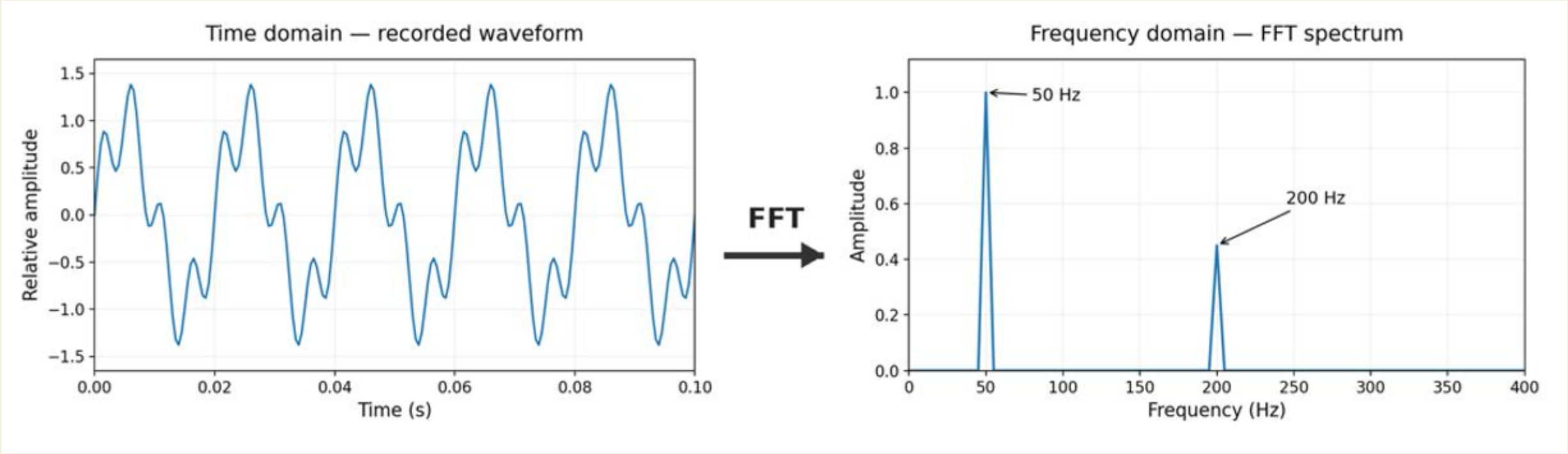
A sound signal can contain several frequencies simultaneously, producing a complex waveform.

Time domain: Shows how the amplitude of a signal changes over time.

Frequency domain: Shows which frequencies are present in the signal and how strongly each contributes.

For example, a signal containing two frequencies (50 Hz and 200 Hz) may appear as a complex waveform in the time domain, but as two distinct peaks in its frequency spectrum.

The frequency domain helps us identify the frequency components hidden within a complex signal.



Fourier Transform and FFT

The Fourier Transform is the mathematical tool that decomposes a signal into its frequency components.

Conceptually:

Time series

↓

Fourier Transform

↓

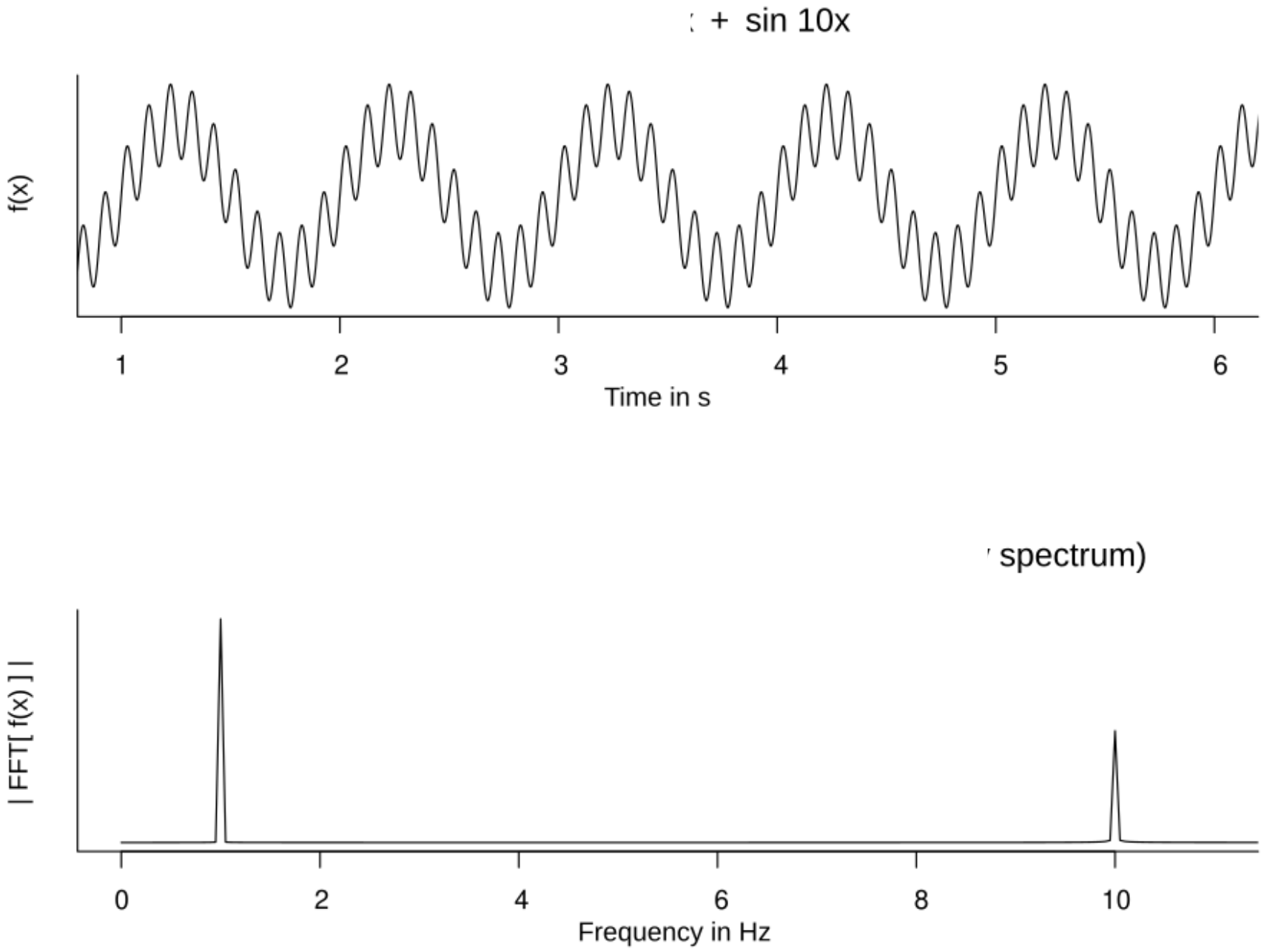
Frequency spectrum

For digital data, the Fast Fourier Transform (**FFT**) is an efficient algorithm used to calculate the discrete Fourier transform.

You do not need to calculate the mathematics by hand, Python will do this for us.

Top — Time domain
A signal formed by combining two sine waves with frequencies of 1 Hz and 10 Hz.
Both frequency components are present simultaneously, producing a more complex waveform.

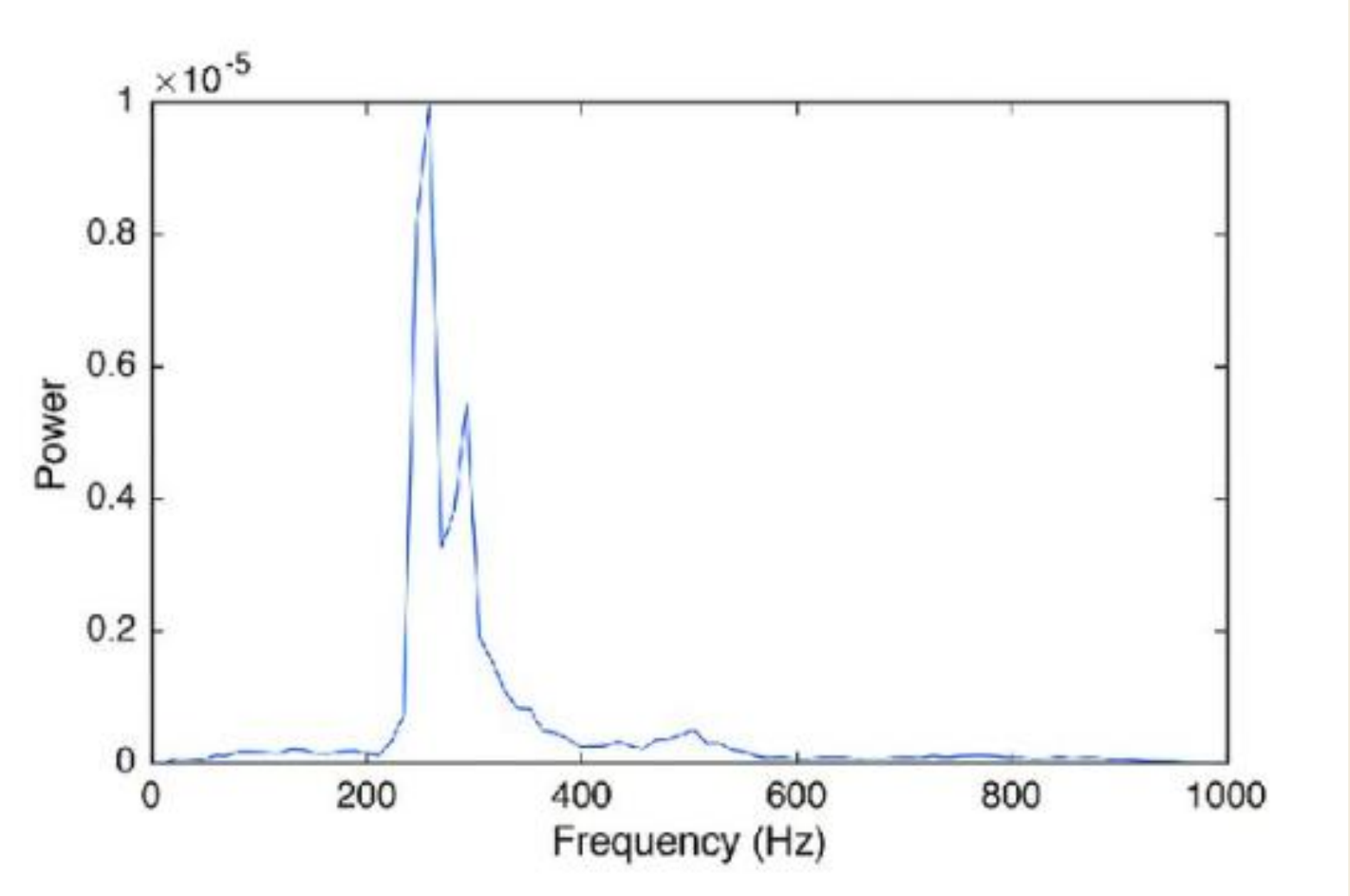
Bottom — Frequency domain
The Fourier Transform reveals two distinct peaks at 1 Hz and 10 Hz, corresponding to the frequencies present in the original signal.



Reading a frequency spectrum

A frequency spectrum shows the strength of different frequency components in a signal.

- x-axis: Frequency, measured in hertz (Hz).
- y-axis: Strength of each frequency component. This may be represented using spectral amplitude, power or power spectral density, depending on the analysis.
- Peaks: Frequencies that contribute strongly to the recorded signal.



Question

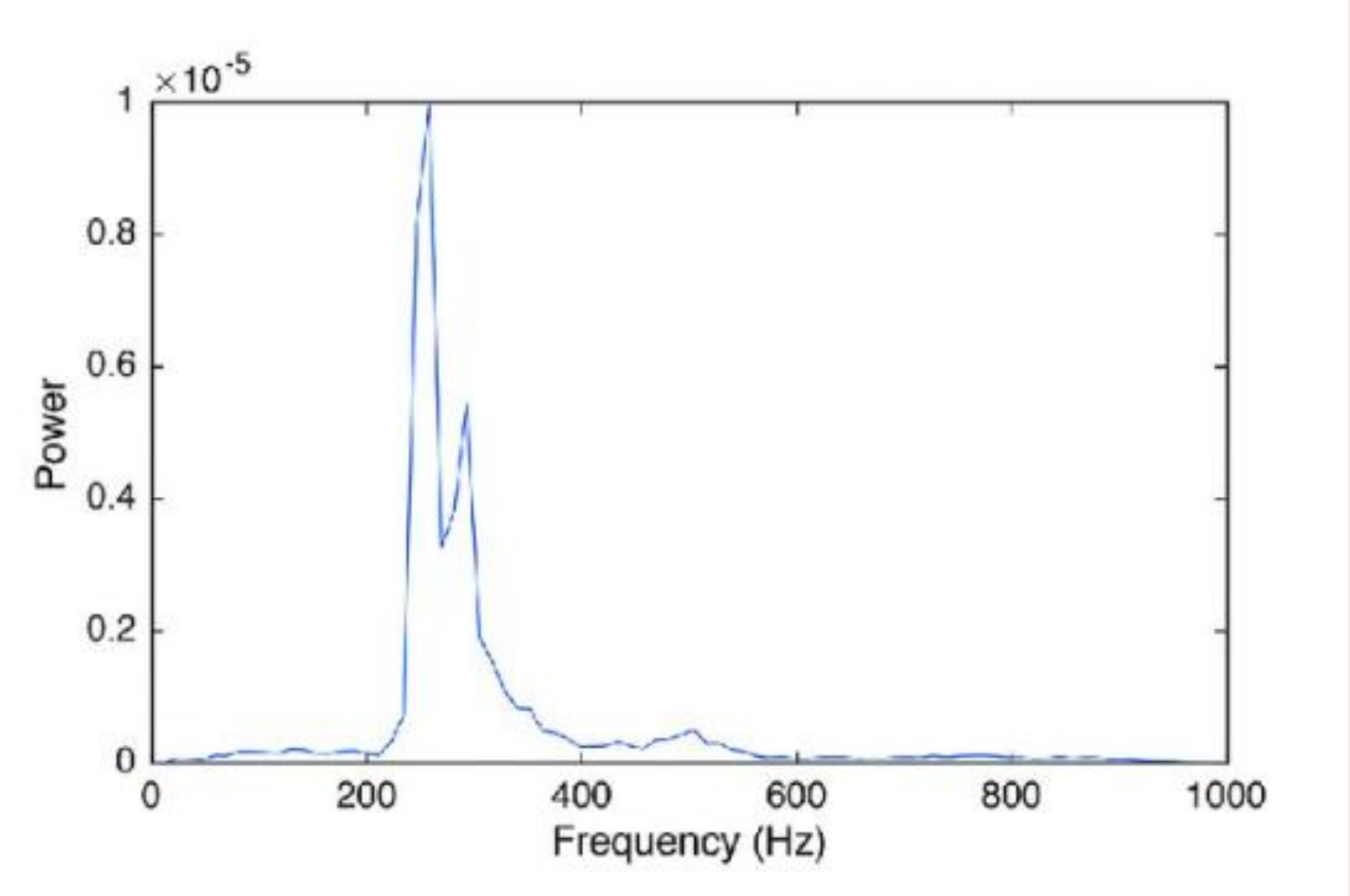
Which frequencies dominate this signal?

Reading a frequency spectrum

A frequency spectrum shows the strength of different frequency components in a signal.

- **x-axis:** frequency
- **y-axis:** amplitude or spectral power

Peaks in the spectrum indicate frequencies that contribute strongly to the signal.



Question

Which frequencies dominate this signal?

Answer: 260 Hz and 290 Hz

Why filter an acoustic signal?

Filters are used to retain selected frequency ranges and reduce unwanted parts of a signal. They can be used to:

- reduce unwanted noise
- isolate a frequency range of interest
- focus on a particular sound source

Common filters

Low-pass filter

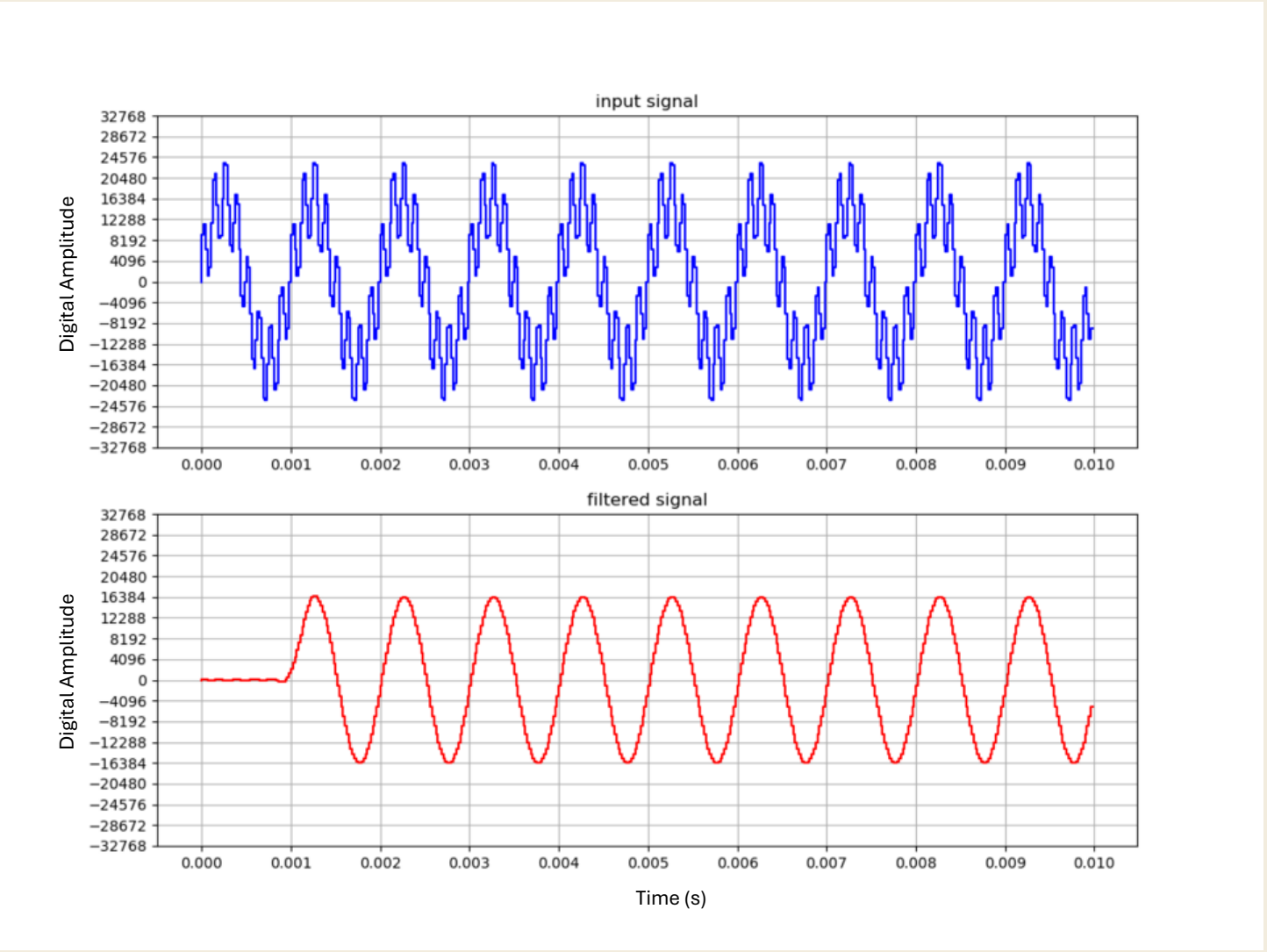
Keeps lower frequencies and reduces higher frequencies.

High-pass filter

Keeps higher frequencies and reduces lower frequencies.

Band-pass filter

Keeps frequencies within a selected range.



Example of low-pass filtering: High-frequency oscillations are attenuated, revealing the dominant low-frequency component.

Which filter would you choose?

1. Reduce high-frequency noise while keeping lower frequencies

2. Isolate a whale call mainly between 20–100 Hz

3. Remove very low-frequency drift

4. Isolate high-frequency dolphin clicks

Which filter would you choose?

1. Reduce high-frequency noise while keeping lower frequencies
   Low-pass

2. Isolate a whale call mainly between 20–100 Hz
   Band-pass

3. Remove very low-frequency drift
   High-pass

4. Isolate high-frequency dolphin clicks
   High- pass

Why is a frequency spectrum not always enough?

A frequency spectrum tells us:

which frequencies are present

but not necessarily:

when those frequencies occur

Two signals can contain similar frequencies but have very different timing.

For changing or intermittent sounds, we need a representation that includes both:

- time
- frequency

FFT tells us WHAT frequencies are present…
…but not WHEN they occurred.

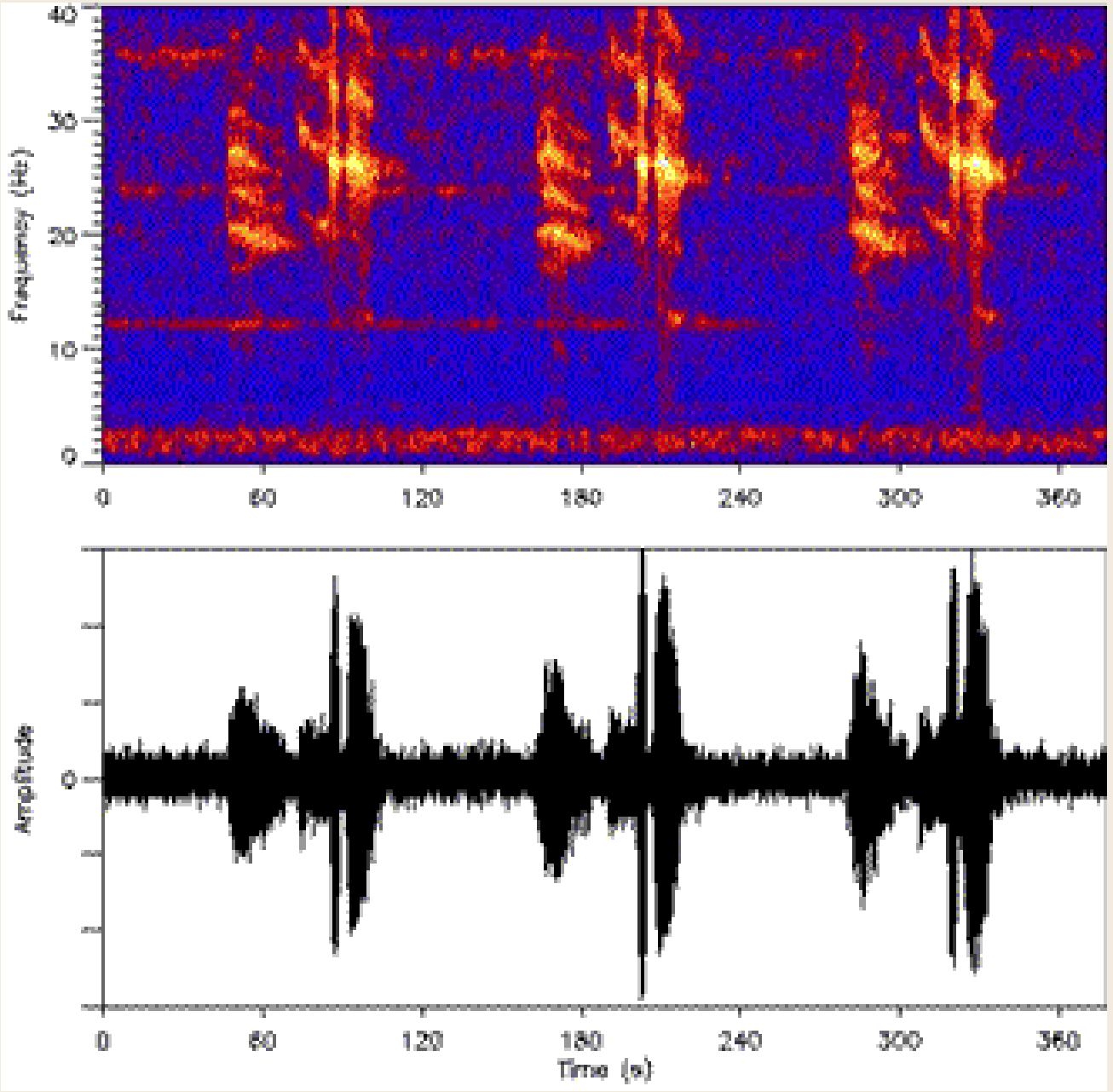
The spectrogram

A spectrogram shows how the frequency content of a signal changes through time.

- **x-axis:** time
- **y-axis:** frequency
- **colour or intensity:** signal strength

A spectrogram can be thought of as a sequence of short frequency analyses applied through time.

It combines information from the: **time domain + frequency domain**

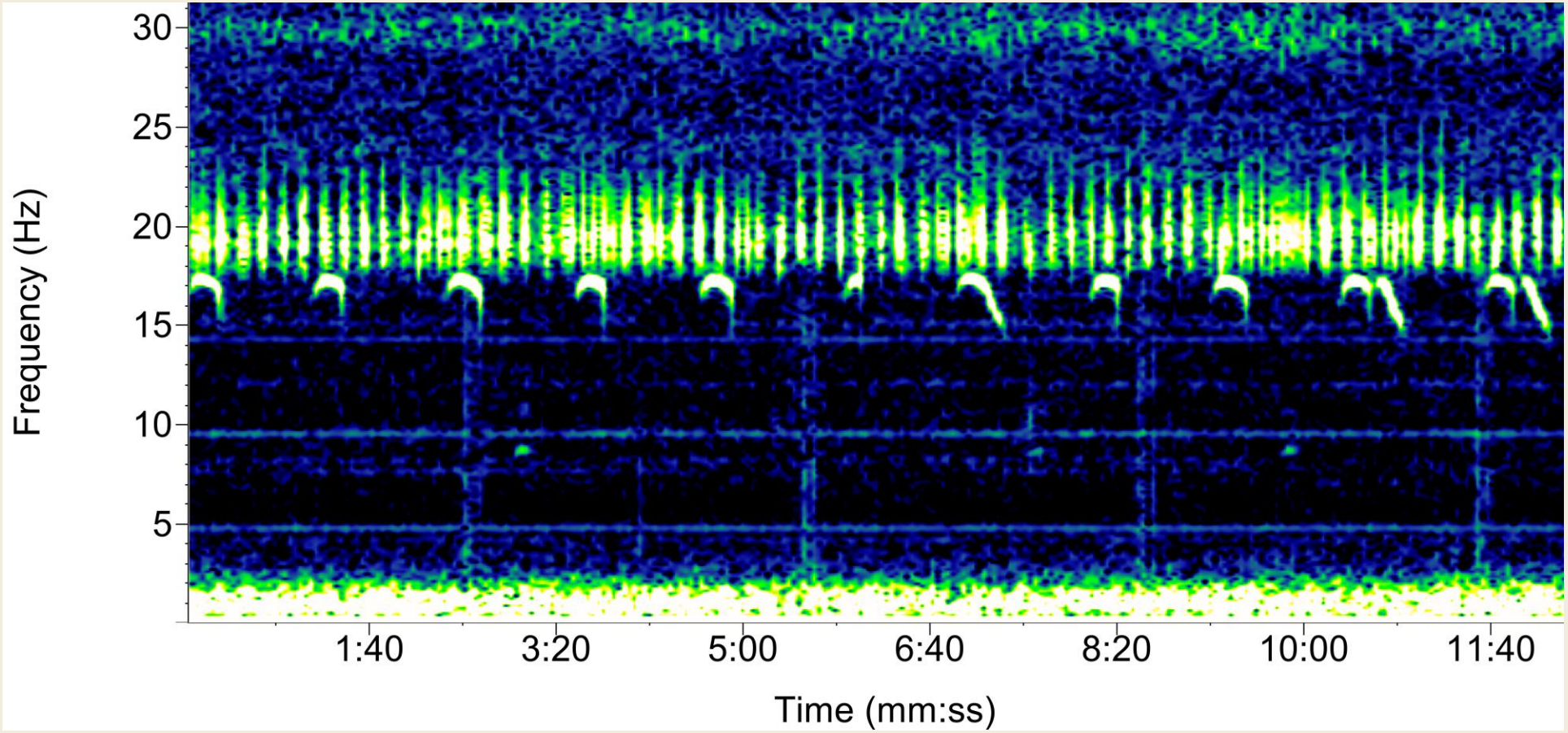


https://www.pmel.noaa.gov/acoustics/specs_all.html

NOAA FISHERIES

More examples at:
https://www.fisheries.noaa.gov/national/science-data/sounds-ocean-mammals





Blue whale low frequency sound.

Introduction to Python and Jupyter Notebook

Throughout this teaching package, we will use Python and Jupyter Notebook to explore and analyse real underwater sound recordings.

You will learn how to:

- Load and visualise hydrophone recordings.
- Analyse sound frequencies using the Fast Fourier Transform (FFT).
- Apply filters to isolate frequency ranges of interest.
- Generate and interpret spectrograms.
- Identify characteristics of different underwater sound sources.

What are Python and Jupyter Notebook?

Python is a free, open-source programming language widely used for scientific data analysis and visualisation.
Jupyter Notebook is an interactive environment where we can combine Python code, explanations and figures in a single document (.ipynb).

How does a notebook work?

1. Open the notebook provided with the teaching materials.
2. Read the instructions and execute code cells using Shift + Enter.
3. Explore the resulting figures and answer the questions.

No previous programming experience is required. The notebooks will guide you step by step.

Installing Python and Jupyter Notebook

Step 1 — Install Python

Download and install Python from the official website:
https://www.python.org/downloads/
Follow the installation instructions for your operating system.

Step 2 — Install Jupyter and scientific libraries

Open Command Prompt (Windows) or Terminal (macOS) and enter:
Windows:

```
python -m pip install notebook numpy scipy matplotlib pandas
```

macOS:

```
python3 -m pip install notebook numpy scipy matplotlib pandas
```

Important (Windows): If py is not recognised, replace py with python in any command.

Step 3 — Open your notebook

Save the provided .ipynb notebook and associated data files in the same folder.
Open Command Prompt or Terminal in that folder, then run:

Windows: python -m notebook
macOS: python3 -m notebook

Jupyter will open in your web browser. Select the notebook file to begin.

Tip: Execute cells from top to bottom using Shift + Enter.

Notebook 1 — Signal Processing Basics

In the practical activity, you will analyse a real hydroacoustic recording in Python. Workflow:

1. Load the audio recording

↓

2. Inspect the waveform

↓

3. Calculate the FFT

↓

4. Apply a frequency filter

↓

5. Calculate a spectrogram

↓

6. Interpret the acoustic signal

Take-home messages

- Sampling frequency controls which frequencies can be represented in digital data.
- The Nyquist frequency is half the sampling frequency.
- A waveform shows how a signal changes through time.
- The FFT reveals the frequency content of a signal.
- Filters isolate selected frequency ranges.
- A spectrogram shows how frequency content changes through time.

References

Scientific and educational resources

- SciPy. *Signal Processing (scipy.signal): User Guide*. https://docs.scipy.org/doc/scipy/tutorial/signal.html
- NOAA Pacific Marine Environmental Laboratory (PMEL). *Hydroacoustic Spectrograms*. https://www.pmel.noaa.gov/acoustics/specs_all.html
- NOAA Fisheries (2023). *Sounds in the Ocean: Mammals*. https://www.fisheries.noaa.gov/national/science-data/sounds-ocean-mammals

Figure and media credits

- Slide 6: Signal aliasing demonstration. Laurens R. Krol (2020), Wikimedia Commons, CC0 1.0. https://commons.wikimedia.org/wiki/File:Signal_aliasing_demonstration.svg
- Slide 20: Hydroacoustic spectrogram and waveform. NOAA Pacific Marine Environmental Laboratory (PMEL).
- Slide 21: Blue whale photograph. NOAA Fisheries, Northeast Fisheries Science Center, Pete Duley (2023).
- Slide 21: Blue whale spectrogram. NOAA Fisheries, Northeast Fisheries Science Center, Passive Acoustics Branch (2023).